Credit Risk | Five practice questions

LECTURE 11 · FINA 402 · Luis Ceballos

Try the questions first. Worked solutions begin after all five questions. Use annual rates; semiannual compounding where specified. Other spread premiums are zero in the PD/recovery exercises.

1. Measure the spread

A corporate bond has YTM 6.20%; a comparable Treasury yields 4.70%. What is the credit spread?

A. 15 bp **B.** 150 bp **C.** 620 bp **D.** 1,090 bp

2. Default and recovery

Annual PD is 3%, recovery is 40%, and the comparable Treasury yields 4%. What spread and required YTM does our approximation imply?

A. 120 bp; 5.20% **B.** 180 bp; 5.80% **C.** 300 bp; 7.00% **D.** 180 bp; 1.80%

3. A change in recovery

Keep annual PD at 3% and Treasury YTM at 4%. Recovery rises from 40% to 60%. What is the new spread, and how does required YTM change?

A. 180 bp; unchanged **B.** 120 bp; falls by 60 bp **C.** 60 bp; falls by 120 bp **D.** 240 bp; rises by 60 bp

4. Price at issuance

FV = $1,000; annual coupon = 6%; maturity = 2 years; semiannual payments. Treasury YTM is 4%, annual PD is 2%, and recovery is 50%. What is the initial price and trading classification?

A. $981.41; discount **B.** $1,000.00; par **C.** $1,018.81; premium **D.** $1,037.72; premium

5. Five-bond portfolio

Invest 20% of market value in each bond. All have the same 4% Treasury benchmark. The annual PD / recovery pairs are A: 1% / 60%; B: 1.5% / 60%; C: 2% / 55%; D: 3% / 60%; E: 3.5% / 60%. What are the portfolio weighted-average spread and YTM?

A. 450 bp; 8.50% **B.** 90 bp; 4.90% **C.** 110 bp; 5.10% **D.** 90 bp; 0.90%

Worked solutions

1. Measure the spread - Answer B

I subtract the Treasury YTM from the corporate YTM. The difference is 1.50 percentage points; each percentage point is 100 basis points.

$$s = 6.20\% - 4.70\% = 1.50\% = 150\ bp$$

2. Default and recovery - Answer B

Step 1. A 40% recovery implies a 60% loss if default occurs. Apply the annual PD to that loss fraction.

$$s = 0.03(1 - 0.40) = 0.018 = 1.80\% = 180\ bp$$

Step 2. Add compensation for credit risk

The government yield compensates for the benchmark investment. The spread is additional, so add 1.80 percentage points to 4%.

$$y = 4.00\% + 1.80\% = 5.80\%$$

3. A change in recovery - Answer B

Step 1. Nothing changes about the chance of default. What changes is the loss after default: it falls from 60% to 40%.

$$s_{new} = 0.03(1 - 0.60) = 0.012 = 1.20\% = 120\ bp$$

Step 2. Recompute YTM and compare

The new required YTM is 5.20%. Compare it with the original 5.80%, not with the coupon rate.

$$y_{new} = 4\% + 1.20\% = 5.20\%, \quad \Delta y = 5.20\% - 5.80\% = -60\ bp$$

Higher recovery reduces the required spread. With unchanged promised payments, the lower required yield would raise the bond price.

Price the promised payments

4. Price at issuance - Answer C

Step 1. Compute the spread and annual required yield before discounting.

$$s = 0.02(1 - 0.50) = 0.01 = 100\ bp, \qquad y = 4\% + 1\% = 5\%$$

Step 2. Map the cash flows

The coupon is based on face value, not price. Each half-year pays $30. There are four payments, and principal is included only in the final payment.

$$C = \frac{1{,}000 \times 0.06}{2} = 30, \qquad N = 2 \times 2 = 4$$

Time (years)	0.5	1.0	1.5	2.0
Promised cash flow	$30	$30	$30	$1,030

Step 3. Discount at the six-month rate

The annual quoted YTM is 5%, so the periodic rate is 2.5%. Keep the maturity payment separate from the earlier coupons.

$$P_0 = \frac{30}{1.025} + \frac{30}{1.025^2} + \frac{30}{1.025^3} + \frac{1{,}030}{1.025^4} = 1{,}018.81$$

Excel: `=-PV(5%/2,2*2,30,1000)`. The price is a premium because the 6% coupon exceeds the 5% required YTM. The coupon rate determines payments; it is not the discount rate.

Combine the five bonds

5. Five-bond portfolio - Answer B

Step 1. Compute each spread separately. For Bond C, a 55% recovery means a 45% loss fraction.

$$s_C = 0.02(1 - 0.55) = 0.009 = 90\ bp$$

Bond	Annual PD	Recovery	Spread (bp)	YTM
A	1.0%	60%	40	4.40%
B	1.5%	60%	60	4.60%
C	2.0%	55%	90	4.90%
D	3.0%	60%	120	5.20%
E	3.5%	60%	140	5.40%

Step 2. Average using investment weights

Each bond has 20% of market value. Add the five spreads and multiply by 0.20. Adding them without the weight gives five times the correct spread.

$$\bar{s} = 0.20(40 + 60 + 90 + 120 + 140) = 90\ bp$$

Step 3. Average the required yields

We can average the five yields directly. Because the Treasury benchmark is common, we can also add the average spread to 4%. Both routes agree.

$$\bar{y} = 0.20(4.40 + 4.60 + 4.90 + 5.20 + 5.40) = 4.90\%$$

This is the portfolio weighted-average YTM requested in the question. It is not a calculation of the exact IRR of the combined cash flows, and it is not a guaranteed realized return.